CHAPTER 2: THE DATABASE ENVIRONMENT

Activity Title; Designing a Database Environment for a University Student Management System

1. COMPETENCY TO BE DEVELOPED

By the end of the activity, the student should be able to:

Analyse a real-world database environment and design an appropriate database architecture and data model while applying ANSI-SPARC architecture, data independence, relational database concepts and the Database SDLC.

PART A: REAL-WORLD SCENARIO

Scenario: UCU Student Management System

Uganda Christian University wants to develop an integrated Student Management System.

Currently, different offices manage different categories of information:

- Admissions manages admission information.
- Registry manages student academic records.
- Finance manages fees information.
- Lecturers manage course and assessment information.
- Students need to view their registration and academic information.
- Management requires reports for decision-making.

The university wants these different users to access information from **one database**.

However, management is concerned about security and database organization.

For example:

- Should a student see another student's information?
- Should a lecturer see a student's financial balance?
- Should Finance see examination marks?
- What happens if the physical storage of the database changes?
- What happens if the university changes part of the database structure?

The university has therefore hired your group as **Database Consultants**.

PART B: ACTIVITY 1 — IDENTIFY DATABASE USERS

Student Task

Identify at least six users of the proposed system.

Complete:

User	Information Required	Should See Entire Database?	Reason
Student			
Lecturer			
Finance Officer			
Registry Officer			
Admissions Officer			
Management			

Arrange the following into the correct ANSI-SPARC levels:

- Student View
- Lecturer View
- Finance View
- Complete logical structure of database
- Student entity
- Programme entity
- Registration entity
- Relationships between entities
- Constraints
- Physical storage
- File organization

- Storage devices

EXTERNAL LEVEL????

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CONCEPTUAL LEVEL????

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INTERNAL LEVEL????

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PART D: ACTIVITY 3 — DESIGN USER VIEWS

Student Task

Design appropriate views for:

1. Student

2. Lecturer

3. Finance Officer

Should the lecturer see the student's financial balance?

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PART E: ACTIVITY 4 — DATA INDEPENDENCE

The DBA announces:

"The physical storage structure of the database will be changed to improve performance."

Question

Should the student application necessarily be rewritten?

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Task: Identify the Type of Data Independence

Situation	**Answer**
Physical file organization is changed	
New attribute is added to the conceptual database structure	
Storage device is changed	

PART F: ACTIVITY 5 — SELECT A MULTI-USER ARCHITECTURE

The university is considering:

1. Teleprocessing
2. File Server
3. Client-Server

Compare the three architectures.

Architecture	**Description**	**Advantage**	**Disadvantage**	**Suitability**
Teleprocessing				
File Server				

Architecture	Description	Advantage	Disadvantage	Suitability
Client-Server				

PART G: ACTIVITY 6 — SELECT A DATA MODEL

The three models presented in the lecture are:

- Hierarchical
- Network
- Relational

Task

With reason , Which model should the university use?

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PART H: ACTIVITY 7 — BUILD THE RELATIONAL MODEL

Student Task

Identify table, attributes and keys for a table named STUDENT :

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PART I: UNDERSTANDING RELATIONAL TERMINOLOGY

Example

STUDENT

StudentID | FirstName | LastName | ProgrammeID

S001 | John | Okello | BIT01

S002 | Mary | Namara | BIT02

S003 | Peter | Ouma | BIT01

Questions

1. **What is the relation?**

2. What is an attribute?

3.What is a tuple?

4. What is the primary key? Why

PART J: ACTIVITY 8 — CONCEPTUAL, LOGICAL AND PHYSICAL MODELS

Stage 1: Conceptual Model

Student Task

Base on the case study to identify the state the Conceptual Model :

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Stage 2: Logical Model

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Stage 3: Physical Model

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